

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No-12	Team No- 9	Project Title- Online Voting System with Aadhaar-Style OTP Verification
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1. Abstract

Online Voting System with Aadhaar Style OTP Verification is a secure, efficient, and user-friendly web-based application developed to modernize the traditional voting process. The main objective of the system is to provide voters with a convenient platform for casting their votes online while maintaining voter authentication, vote integrity, and controlled access to the election process. The system is designed to reduce manual effort, minimize paperwork, and provide faster election management and result generation.

The proposed system incorporates OTP-based voter verification, inspired by the authentication approach used in Aadhaar-enabled services. During the voting process, a registered voter provides the required identification information, after which a One-Time Password (OTP) is generated and sent to the voter's registered mobile number. The voter must enter the correct OTP within the permitted time to successfully authenticate their identity. Only verified voters are allowed to proceed to the voting interface.

After authentic action, the voter can view the list of available candidates and select one candidate to cast the vote. The system maintains a record of the voting status to ensure that a voter cannot cast multiple votes in the same election. Candidate information, voter records, election details, OTP verification data, and voting records are managed through a centralized database. An administrator module is included to manage voter registration, candidate details, election information, and voting results.

The system can also provide real-time or final result generation based on the votes stored in the database. Proper validation and access control mechanisms are considered to improve the security and reliability of the application. Overall, the project demonstrates the practical application of database systems, backend development, OTP authentication, user management, and secure voting workflows to create a structured online voting platform. The proposed system

aims to make the voting process more accessible and efficient while maintaining essential authentication and voting controls.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090132	Keerthana. G	
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3. Frameworks / Platforms / Software Used

Frontend: HTML, CSS, JavaScript

Backend: Java / Spring Boot

Database: MySQL

Authentication: OTP-based verification

API Testing: Postman

Development Environment: Visual Studio Code / IntelliJ IDEA

Database Management: MySQL Workbench

Version Control: Git and GitHub

Guide Name and Signature (remarks if any) _____